

International AS and A-level

Chemistry (9620)

CH03 Inorganic 2 and Physical 2

Report on the examination

June 2024

REPORT ON EXAMINATION: INTERNATIONAL A-LEVEL CHEMISTRY (9620) CH03 JUNE 2024

This examination of the international specification highlighted some key general points about students' performance that may prove useful.

Students need to:

learn the meanings of key terms accurately.

ensure that chemical equations are balanced for both atoms and charge.

show all working with a clear explanation for each step of their calculation.

be familiar with units and be able to use them to calculate quantities,

understand that in aqueous solution the hexa-aqua ion $[M(H_2O)_6]^{2+}$ or $[M(H_2O)_6]^{3+}$ is present.

understand how to give conventional cell representations for electrochemical cells including the use of lines and commas to separate species.

understand how electrode potentials can be used to predict redox reactions.

be very familiar with all the practical procedures contained in the specification, and very importantly understand why they carry out the procedures that they do in practical work, rather than knowing just what to do.

QUESTION 01: The enthalpy of lattice dissociation of calcium bromide

01.1 The definition for the enthalpy of lattice dissociation was well known, although students did not always state that one mole was needed or that the compound should be ionic. Some students didn't state that gaseous ions would be formed.

01.2 The calculation was well completed with about two-thirds of students scoring both marks. The most common error was not appreciating that two moles of bromide ions are hydrated and so the value for the enthalpy of hydration of bromide ions was not always multiplied by two.

01.2 Many students realised that the reason for the difference between the theoretical and experimental values for the enthalpy of lattice dissociation was due to covalent character. A common incorrect approach was to discuss heat loss.

01.3 This calculation was well completed, although only about one third of students realised that the enthalpy change was exothermic so that the final value should be negative. Some students did not realise that because the density of the solution is 1.00 g cm^{-3} then the mass of solution would be the same, in grams, as the volume, in cm^3

01.4 About half of the students understood that adding the washings meant that the student did not know the actual volume or mass of the solution that would be present in the plastic cup.

QUESTION 02: The Born–Haber cycle for barium oxide

02.1 This was very well completed with about three-quarters of students scoring all three marks. Some students did not include state symbols.

02.2 This also was very well completed and just over three-quarters of students were able to calculate the enthalpy of formation. Some students halved the enthalpy of atomisation of oxygen.

QUESTION 03: The reaction of magnesium with carbon dioxide

03.1 The graph was well drawn with the majority of students scoring both marks. Some students misplotted one or more of the points, whilst others tried to force their line of best fit through the corner point (0, –220).

03.2 Most students understood that the value for the enthalpy change could be deduced from the y-intercept.

03.3 The majority of students (about 80%) realised that the entropy change could be determined from the gradient of the line of best fit. However, only a few students (just over a third) understood that the gradient was equal to $-\Delta S$.

03.4 This question was more challenging and only about one third of students understood that a change in gradient meant that the entropy change was different, so this would lead to a change of state of one of the reactants or products.

QUESTION 04: The standard electrode potential for the $\text{MnO}_4^-/\text{Mn}^{2+}$ electrode

04.1 Many students knew the conditions for the standard hydrogen electrode, but a large number of students stated that the gas should be at 1 atm pressure rather than 100 kPa

04.2 Only about one quarter of students scored both marks. Students should only include ions or molecules in the cell representation (rather than the formulas of ionic compounds such as KMnO_4). Only a small number of students used lines (to separate species in different phases) and commas (to separate species in the same phase) correctly. In addition students often gave the species the wrong way around so that the oxidised species were not adjacent to the salt bridge.

04.3 Students did not always make reference to the data from Table 4 when explaining why sodium chloride should not be used in the salt bridge. Only about one third of students could explain that the chloride ions would be oxidised by the manganate(VII) ions.

04.4 This was well completed with most students (about three quarters) correctly calculating the standard electrode potential of the $\text{Fe}^{3+}/\text{Fe}^{2+}$ electrode.

04.5 About half of the students correctly deduced the equation for the reaction when the cell is being recharged. Many students gave the equation for the reaction when the cell is being discharged.

04.6 Just over half the students realised that any water would react with the lithium in the cell.

QUESTION 05: Acids and bases

05.1 Nearly all the students correctly calculated the pH of the hydrochloric acid and gave this to two decimal places.

05.2 The expression for K_w was given accurately by most students, although some students divided by the concentration of water.

05.3 This was also well answered, and the majority of students scored both marks for the calculation of the pH of the aqueous solution of sodium hydroxide. Some students used the value of K_w at 25 °C rather than the value of K_w at 30 °C.

05.4 Only a small number of students correctly explained why the pH of the aqueous solution of sodium hydroxide decreased when the solution is heated. Students did not realise that the dissociation of water is endothermic. Some students stated that the concentration of hydroxide ions changed.

05.5 About half of students scored all the marks on this calculation. Some students used the incorrect volume in their calculation. The volume at the end point can be determined from the vertical section of the pH curve in Figure 4.

05.6 Some students did not correctly determine the volume at half equivalence, and instead halved the volume shown by the end of the curve on Figure 4. The majority of students were able to correctly calculate K_a from the pH.

QUESTION 06: Metal ions and complexes

06.1 Most students were able to give the full electron configuration of Fe^{2+} , although some students gave the electron configuration of the atom rather than the ion. A further common error was to give an electron configuration that ends $3d^54s^1$.

06.2 This was not well answered with students not stating that many atoms were each donating electron pairs and rather just stated that many coordinate bonds were formed.

06.3 The energy change was well calculated, and the majority of students scored both marks. A few students did not correctly convert the wavelength into metres for the calculation.

06.4 The majority of students made some progress on this explanation. Students generally understood that the changing number of particles in the equation can be used to provide an indication about the entropy change for the reaction but then didn't always link this to the Gibbs free-energy change.

06.5 About one fifth of students were able to describe a test-tube reaction that showed that the Ca^{2+} ions had been removed from the aqueous sample. Many students gave an observation which would show that Ca^{2+} ions were present rather than absent.

QUESTION 07: Transition metals

07.1 Only about one quarter of students were able to state why zinc is not a transition metal. Students did not state that zinc ions have an incomplete d sub-shell and often only just focused on atoms of zinc.

07.2 About two thirds of students scored all three marks. Students were able to write two equations to show how V_2O_5 act as a catalyst in the Contact process. However, explanations about why V_2O_5 is heterogenous. Students need to compare the states/phase of the catalyst with the state/phase of the reactants.

07.3 Most students understood that impurities would poison the catalyst by blocking the active sites.

07.4 This question was found to be challenging by students. Students did not appreciate that the electrode potential for the reduction of Zn^{2+} should be more negative than the electrode potential for the reduction of V^{3+} to V^{2+} and that it should also be more positive than the electrode potential for the reduction of V^{2+} to V. The overall equation was not always balanced for atoms or charge and electrons were often included in this equation.

07.5 Most students were able to correctly identify the shape of the four transition metal complexes.

QUESTION 08: Aluminium

08.1 The majority of students identified the p block as that which contains aluminium.

08.2 The equation for the reaction of aluminium with oxygen was correctly completed by the majority of students.

08.3 Equations for the reaction of aluminium hydroxide with an acid and with an alkali where often not balanced. A number of students wrote an equation for the reaction of aluminium hydroxide with hydrochloric acid to form aluminium chloride. As aluminium chloride would hydrolyse in aqueous conditions this would not form in this reaction.

08.4 About three quarters of students correctly showed covalent and co-ordinate bonds in Al_2Cl_6 . However, co-ordinate bonds were sometimes pointing in the wrong direction.

08.5 This question was found to be challenging by students. Student did not always give an equation within their answer and should have written the acidity reaction. Students should appreciate that the high charge density of the aluminium ion affects the O–H bonds in the water ligands (by polarising them and so weakening the bond).

08.6 The observations for this reaction were well known and the majority of students scored both marks for these. However, the equation for the reaction of $[Al(H_2O)_6]^{3+}$ ions with carbonate ions was less well known and was often not always balanced.

QUESTION 09: Redox titration

09.1 The mean titre was calculated correct by the majority of students, although some included all three titre values rather than just those that were within 0.10 cm^3 of each other (concordant).

09.2 About two-thirds of students correctly deduce the number of moles of $\text{S}_2\text{O}_3^{2-}$ required to react with the iodine produced from the reaction of one mole of IO_3^- ions.

09.3 This calculation was well completed by most students with just over half scoring all four marks. Students did not always appreciate that only 25.0 cm^3 of the KIO_3 was used in each titration from the stock solution of 250 cm^3 and so did not scale their quantities appropriately.

09.4 Only about one third of students correctly explained why excess potassium iodide and sulfuric acid are used. Students tended to focus on ensuring that all the thiosulfate reacts rather than all the IO_3^- ions reacted.

09.5 Just over one quarter of students scored the mark. Many students incorrectly discussed colour changes rather than understanding that because these substances are used in excess the exact amounts of potassium iodide and sulfuric acid do not need to be known.